SPECIFICATION

Product Name： IoT Module

Model No.： TI602B3S

Type： URAT Interface

Version： V1.0

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Content

1. Profile

TI602B3S is a WIFI +BLE IOT module. It consists of a highly integrated chip BL602 and several peripheral components, It supports 2.4Ghz and WIFI standard IEEE 802.11 b/g/n. Adopting a 3.3V single power supply, the module can be flexibly applied to various consumer products. This module application for the smart home, household electrical appliances and smart light control.

1.1 Features

Built-in low power 32bit CPU and also can be an application processor

WIFI standard is 802.11 b/g/n

WPS/WEP/WPA/WPA2 Personal/WPA2 Enterprise/WPA3

PWM control

Support timer operation

Support WIFI remote control

Support timer operation

Supports compatible pairing mode/Bluetooth pairing

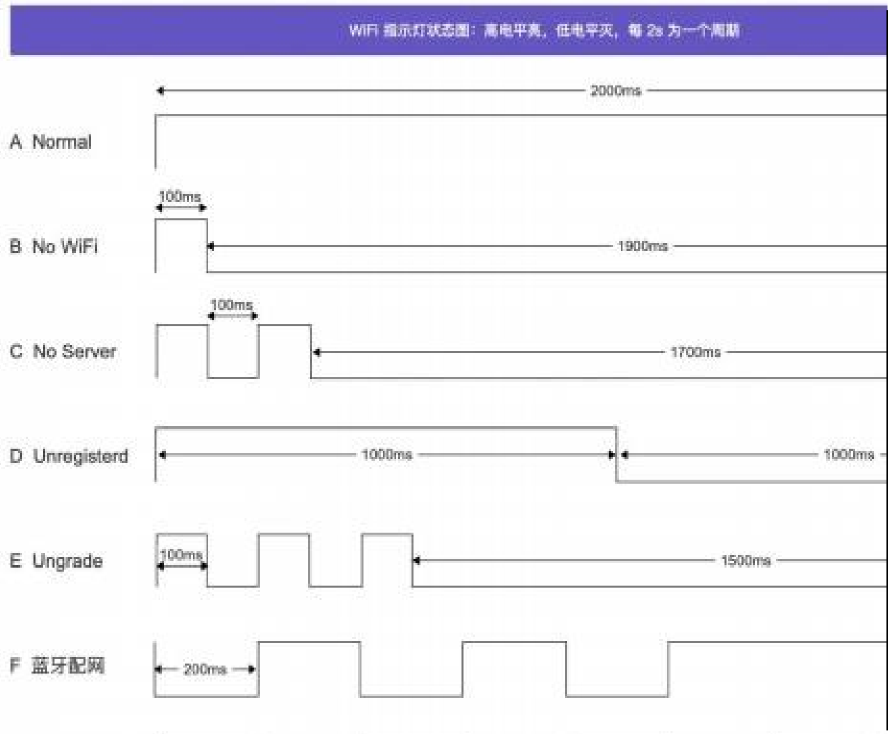
OTA upgrade

LAN control

Support voice control by smart speaker of Alexa, Google assistant , Tmall-genie, Mijia, Xiaodu

2.Function Description

2.1 WIFI indicator light



A: Device on the line

B: Internet connection fail

C: Internet connection successes, but The device is not connected to the server

D: Device is registering

E: The device is upgrading firmware (OTA)

F: BLE pairing

G: AP paring

H: The device is in factory mode

2.2 Pairing network

TI602B3S has two way for paring network: BLE pairing and compatibility pairing.

2.2 Entering pairing network mode

2.2.1 TI602B2S has some pairing networking mode as following:

1. Unpaired devices automatically enter BLE network mode after powering on
2. Pull down the pin with 5s for any button input when the device not under the pairing network mode, the chipset will enter into BLE pairing mode. Pull down the pin with 5s for any button input again when the device under the BLE pairing mode, the chipset will enter into compatibility pairing.
3. When the App remove the device, the device will be reset as factory default and enter into BLE pairing mode.

2.2.2 Exit the pairing network mode

1. The device will exit the pairing network status when the pairing is successful
2. Over 3mins the device will auto exit pairing network status when the device under the pairing network mode.

2.3 Switch control

Pin of switch input defaut is High level, effective when the pull down. When triggered, the relay output pin corresponding to the button flips

2.4 Timer control

TI602B3S Support timer type as following:

1. Timing : Set time for the specified action
2. Delay: Execute the specified action after a certain period of time (The delay time max is 23 hours 59mins)

Attention:

Action function only has turn on /off.

Timing and delay have option for enable and disable

Module support 20 timers, but only support 8 timers run at the same time.

2.5 Electrification reaction

Electrification reaction: The TI602B3S is powering on again after power off, the module can be set as three mode as following:

- Star
- Close
- Restore the state before the power outage

  Factory default as close

2.6 LAN (local area network) Control

App can control the module switch directly when the mobile phone and device are connected under the same local area network without using the cloud platform.

2.7 The indicator light of network

TI602B3S supports configuring network indicator lights through the APP, (light off status when the device online)

2.8 Delete device

User can delete the device by APP, the device remove from the APP, the device auto reset as factory default mode.

3. Electrical Characteristic

3.1 Basic parameters

Test condition: 3.3V±10%，GND=0V ; Temperature 25℃	
Model	TI602B3S
Interface	PWM/IC/GPIO
Working Voltage	3.0V -3.3V
Working Current	Max Working Current:340mA
Working temperature	0℃~45℃
Storage environment	Temp: -10℃~-+75℃ relative humidity 20%RH~80%RH
Wireless network type	WIFI:STA/AP/STA+AP Bluetooth: BLE 5.0
Security	WPS WEP WPA/WPA2 Personal /WPA2 Enterprise /WPA3
Encryption type	WEP64/WEP128/TKIP/AES 128/192/256
Firmware update	OTA Remote upgrade

3.2WIFI parameters

Test condition: 3.3V±10%，GND=0V ; Temperature 25℃	
WIFI Standard	IEEE 802.11b/g/n
Frequency Range	WIFI:2412-2482MHz BT:2.400GHz to 2.4835GHz
Working Voltage	3.0V -3.3V
Transmitting Power	802.11b:±19dBM(@11Mbps) 802.11g:±16dBM(@54Mbps) 802.11n:±16dBMReceiving tivi(@HT20MCS7)
Receiving Sensitivity	802.11b: -92 dBm(@11Mbps, CCK) 802.11g: -76 dBm(@54Mbps, OFDM) 802.11n: -72 dBm)MCS7)
Antenna Type	PCB antenna on board

3.2 WIFI RF parameter

Test condition: 3.3V±10%，GND=0V ; Temperature 25℃				
Item	Min	Typ	Max	Unit
Input frequency	2412	-	2482	MHz
Output impedance	-	50	-	Ω
Input reflection	-	-	-10	dB

3.4 Sensitivity

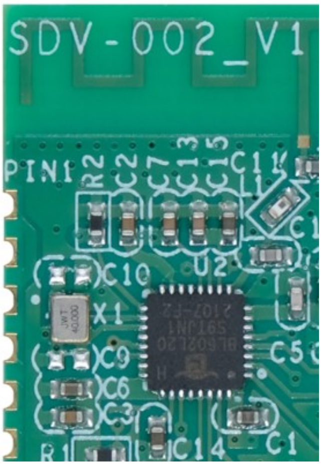
Item	Min	Typ	Max	Unit
CCK 1Mbps		-96		dBm
CCK 11Mbps		-90		dBm
6Mbps(1/2BPSK)		-92		dBm
54Mbps(3/4 64-QAM)		-76		dBm
HT20, MCS7(65Mbps, 72.2Mbps)		-72		dBm

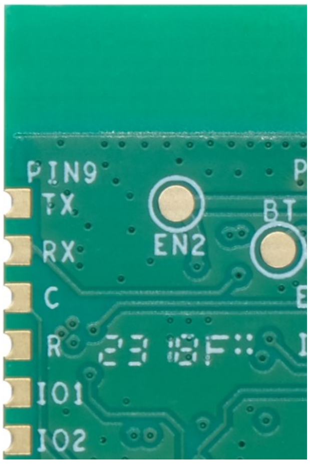
3.5 Power consumption parameters of modules in continuous transmission state

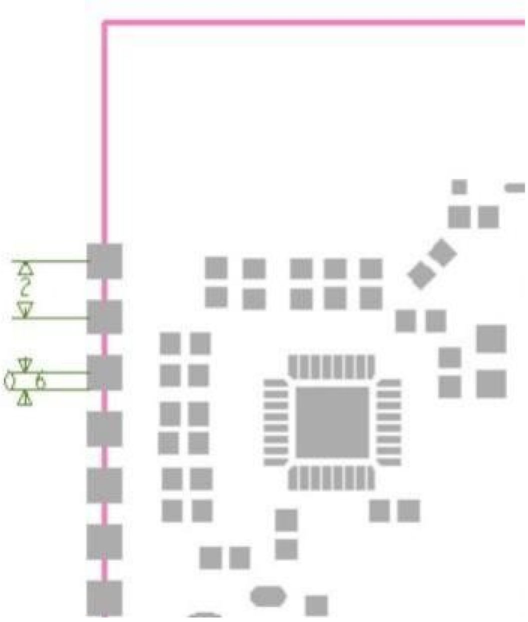
Note: Temperature:25℃, DC:3.3V

Item	Average value	Max	Unit
802 . 1 1b 1Mbps 21 dBm TX	227	308	mA
802 . 1 1b 1Mbps 23 dBm TX	254	356	mA
802 . 1 1n MCS7 17 dBm TX	232	256	mA
802 . 1 1n MCS7 18dBm TX	223	268	mA
802 . 1 1n MCS7 19dBm TX	215	284	mA
802 . 1 1n MCS7 23dBm TX	217	340	mA
802 . 11g 54Mbps 21dBm TX	218	338	mA
802 . 11g 54Mbps 23dBm TX	219	340	mA
Module power on, but stop packet	45		mA

4. Module Size



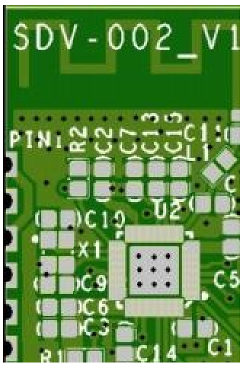


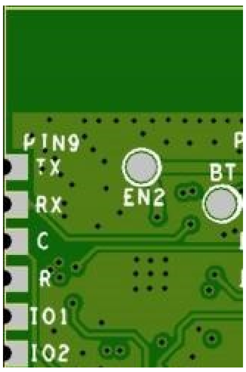


Dimension unit: MM Tolerance: ±0.1MM

Module size: 24mm(L)*16mm(W)*2.1mm(D)

5. Pin Assignment





Pin	Symbol	Function
1	RTS	NC
2	ADC	NC
3	EN	NC
4	IO0	LED indicator: Active low level, it is recommended to connect a 3.3Kx current limiting resistor in series to VCC
5	B	Button 3 input, Active low level. Default as high level
6	G	Button 1 input, Active low level. Default as high level
7	W	Relay control, Active High level
8	VCC	3.3 Power input
9	GND	GND
10	IO3	Relay 2 (Switch control signal output, active high level, default as low level. Turn on as high level.)
11	IO2	Relay 4 (Switch control signal output, active high level, default as low level. Turn on as high level.)
12	IO1	Button 4 input, default as high level. Active low level, pull down over 5s, it will enter into pairing.
13	R	Relay 3 (Switch control signal output, active high level, default as low level. Turn on as high level.)
14	C	Button 2 input, default as high level. Active low level, pull down over 5s, it will enter into pairing.
15	RX	NC
16	TX	NC,

The 5CH PWM of module application for lighting , the pin assignment as following:

Pin	Symbol	Function
1	RTS	NC
2	ADC	NC
3	EN	NC
4	IO0	NC
5	B	Default blue PWM output
6	G	Default green PWM output
7	W	Default warm white PWM output
8	VCC	3.3 Power input
9	GND	GND
10	IO3	NC
11	IO2	NC
12	IO1	NC
13	R	Default red PWM output
14	C	Default cold white PWM output
15	RX	NC
16	TX	NC,

6. Reference design for PCBA

Attention of PCB layout and Module

1. When laying out a PCB, pay attention to the placement of the modules and try to stay as far away from interference sources as possible: magnetic components (such as motors, inductors, transformers, etc.), high-frequency signal devices (such as crystal oscillators, high-frequency clock signals, etc.).
2. Clear space is required in the module PCB antenna area and the 15mm external expansion area (copper laying, wiring, and component placement are strictly prohibited)
3. The capacitance of the power supply (VCC) pin of the module and the capacitance and resistance of other pins of the module should be placed as close as possible to the module pins, and the wiring path should be short.

7. Environment about Working and Storage

Working Temp	Temperature:-20℃to+85℃
	Relative Humidity:10-90%(non-condensing)
Storage Temp	Temperature:-40℃to+85℃(non-condensing)
	Relative Humidity:10-90%(non-condensing)

8. Package details

- Packed as vacuum reel or tray (TDB)
- Reel color: blue   Carton size: 33.7cm*33.7cm*9.5cm
- 2 packs desiccant, 1 pcs humidity card